Isentia: Meeting customer demand for media intelligence with Speech-to-Text

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About Isentia

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About Isentia

Isentia (ASX:ISD) is a leading integrated media intelligence and insights business in APAC. Isentia blends market-leading monitoring experience with analytics to help the world's biggest brands uncover the whole picture—and act on it. Powered by cutting-edge technology and a team of world-class experts, Isentia's mission is to help businesses leap forward to where only genuine insight can take them. To find out more about how Isentia informs better decisions, please visit www.isentia.com (https://www.isentia.com).

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Industries: Media & Entertainment

Location: Australia

(https://cloud.google.com/kubernetes-engine/),

Google Cloud results

- Scales to support demand peaks with volumes 10x greater than demand troughs
- Enables team to focus on algorithms and products rather than underlying technologies
- Ensures smooth transition to remote working with easily accessible

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Google Cloud (https://cloud.google.com/)

Cloud Speech to Text (https://cloud.google.com/speech/)

Cloud Memorystore (https://cloud.google.com/memorystore/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud Storage (https://cloud.google.com/storage/)

Cloud Functions (https://cloud.google.com/functions/)

Google Workspace (https://workspace.google.com/)

Gmail (https://workspace.google.com/products/gmail/)

Sheets (https://workspace.google.com/products/sheets/)

Slides (https://workspace.google.com/products/slides/)

Google Meet (https://workspace.google.com/products/meet/)

Docs (https://workspace.google.com/products/docs/)

collaboration tools

The vast world of social and traditional media can be a challenge for organizations that need to track emerging stories and posts to identify trends that create opportunities to win customers and generate revenue. Media monitoring solves that problem, and Isentia (https://www.isentia.com) is a leading provider of media intelligence in the Asia-Pacific. The organization has turned to AI and machine learning to meet clients' increasingly sophisticated demands.

Founded by advertising executive Neville Jeffress in Melbourne in 1982, Isentia has grown to become a publicly listed company with more than 900 employees across Australia, New Zealand, and the Asia-Pacific, and a provider of media intelligence services to 3,000+ clients in 18 languages.

“We aim to help customers understand what is happening in the market that impacts them by monitoring television, newspapers, radio stations, social media, websites, and blogs and overlaying insights,” says Paul Russell, Chief Technology Officer at Isentia. “Our monitoring goes far beyond cataloging mentions of brands or executive team members, to the supply of information about what people are saying about a business, industry, or interests in near real time.

“That is extremely important, because the value many of our clients gain is the ability to understand a trend or an event and respond quickly.”

For example, if a client is mentioned in a negative context during a talkback radio show, its teams can determine whether to call in and respond.

Driving KPIs

A number of communications and media relations teams rely on Isentia to drive performance against key indicators. This makes providing an accurate, relevant service extremely important to Isentia's success and means the business has to delve deeply into posts on a growing number of social media services, as well as stories across traditional media.

To meet client requirements and provide a foundation to build powerful media intelligence services covering television and radio, Isentia needed to automate the conversion of broadcasters' speech to text. Isentia reviewed its options and selected Speech-to-Text (/speech-to-text), a Google Cloud (https://cloud.google.com/) API powered by AI technologies. "We selected Speech-to-Text based on accuracy and how easy it is to integrate speech recognition into the applications we were developing," says Paul.

"Overall, Google Cloud has made creating intelligent services much easier with an API that allows us to integrate different types of software in a straightforward way. This means we can focus on building algorithms and functionality, rather than making all the various pieces work together.”

—Paul Russell, Chief Technology Officer, Isentia

Deployment in just three months

With help from Google, Isentia deployed its new TV and radio broadcast solution in just three months. “We capture raw broadcast content from television or radio, either as a digital stream or through capture cards in servers, run it through Speech-to-Text, and overlay a range of AI algorithms to produce our media intelligence products," says Paul.

The Isentia solution uses containerized applications running on Google Kubernetes Engine (/kubernetes-engine), which provides the fast horizontal scaling and stability required to reliably capture high-bandwidth broadcast content from a variety of sources. The "source of truth" to orchestrate the containers is a synchronized Firebase Realtime Database (https://firebase.google.com/docs/database). The captured audio and video is then stored in a secure Cloud Storage (/storage) environment. A caching layer, Memorystore (/memorystore), ensures high throughput and sub-millisecond availability of the transcripts and the media clips.

After running Speech-to-Text over the broadcast content, the results are analyzed and enriched via a range of in-house built functions:

- A boundary detection service that uses AI to identify when one news story ends and the next begins, isolating relevant media items and giving clients immediate access to news segments relevant to them
- An ad filtering service that uses AI to identify and remove advertisements from automated broadcast monitoring
- A music filtering service that uses AI to remove the music that plays before or after many broadcast news items
- A service that matches words in text transcripts to the content in the video player, allowing users

to quickly jump to the section of the media item they want to see

Speech-to-Text also provides a foundation for Isentia's developers to apply machine learning, to add proper nouns, and improve the quality of people, company, and place names. "This is incredibly important to us given we monitor through keywords and undertake entity detection," says Paul.

"The collaboration features of Google Docs, Sheets, Slides, Gmail, and Meet—and the integration between the various products—were far superior to our previous software."

—Paul Russell, Chief Technology Officer, Isentia

Supporting demand peaks

With Speech-to-Text and Google Cloud, Isentia can provide a reliable, real-time service to clients during demand peaks with 10x greater volumes than quiet periods. These peaks occur during prime time television news bulletins and current affairs shows, with volumes dramatically lower late at night and into the early morning. Today Isentia only uses Speech-to-Text as the foundation of English-language AI-powered services, but plans to expand it to languages such as Chinese dialects, Malay, and Tagalog.

“Overall, Google Cloud has made creating intelligent services much easier with an API that allows us to integrate different types of software in a straightforward way,” says Paul. “This means we can focus on building algorithms and functionality, rather than making all the various pieces work together. Using Cloud Functions (/functions) to run code and Google Kubernetes Engine means we did not have to run up servers and install a software package, but more importantly, as they are managed services, we do not have the ongoing distraction of running them day to day.”

“We were very grateful that we made the change to Google Workspace as we would have struggled to manage the shift to remote work without it. We moved our people in about three weeks and because of Google Workspace’s accessibility from the browser, we could focus on the impact of the transition on our other tools.”

—Paul Russell, Chief Technology Officer,

Google Workspace eases pandemic transition

Delivering intelligent media services in a fast-changing environment requires close collaboration to ensure all team members are on the same page. When Isentia's current Chief Executive Officer started two-and-a-half years ago, he built a new executive team that reviewed the business's operations and enabling technologies. "We identified early on that we needed to lift the productivity of our team members by giving them better tools to do their jobs," says Paul.

The business reviewed its options and elected to move to Google Workspace (https://workspace.google.com/). "The collaboration features of Google Docs (https://workspace.google.com/products/docs/), Sheets (https://workspace.google.com/products/sheets/), Slides (https://workspace.google.com/products/slides/), Gmail (https://workspace.google.com/products/gmail/), and Google Meet (https://workspace.google.com/products/meet/)—and the integration between the various products—were far superior to our previous software," says Paul.

The business transitioned its workforce to Google Workspace in 2019 and quickly began experiencing improvements in productivity and collaboration.

These benefits were amplified when the coronavirus pandemic spread and Isentia had to send its geographically distributed workforce to work from home. “We were very grateful that we made the change to Google Workspace as we would have struggled to manage the shift to remote work without it,” says Paul. “We moved our people in about three weeks, and because of Google Workspace’s accessibility from the browser, we could focus on the impact of the transition on our other tools.”